

The Evolution of Exploratory Data Analysis

From Tukey's detective mindset to modern AI-assisted
discovery

Titanic case study • methods comparison • critique • current state of the art



EDA is a philosophy, not just a preprocessing step

Why it comes before formal modeling.

- John Tukey introduced EDA as a way of thinking about data before running confirmatory tests such as p-values or regression models.
- The analyst acts like a detective: inspect the structure, surface important variables, and look for anomalies before trusting a model.
- You are “interviewing” the dataset to see whether it is hiding something important.
- If you skip EDA and jump straight to modeling, you risk a sophisticated model built on fundamentally flawed data.

Mindset

Ask first:
What is the data shape?
What is missing?
What looks suspicious?

Principle

Garbage in, garbage out.
EDA protects later modeling
decisions.

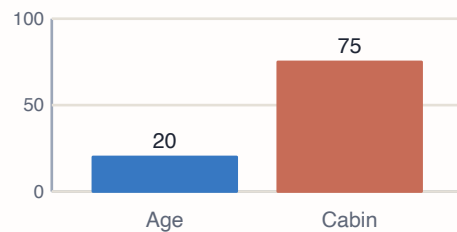
The three core objectives of EDA

Each objective answers a different question before machine learning starts.

1. Data Quality Assessment

- Identify anomalies and outliers that could skew later models.
- Understand missingness, including whether data may be missing systematically.

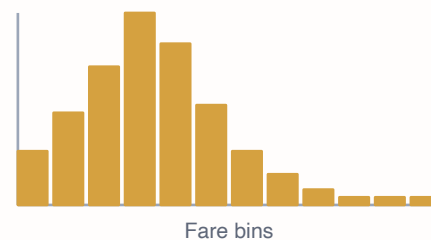
Missingness



2. Univariate Analysis

- Study one variable at a time: center, spread, skew, and unusual shape.
- Use distributions, mean/median/mode, standard deviation, and the IQR.

Distribution shape



3. Bivariate & Multivariate

- Investigate how variables move together using cross-tabs, scatter plots, and correlations.
- Look for relationships that will later guide feature engineering.

Signal map

	Class	Sex	Fare
Survival	82	88	46

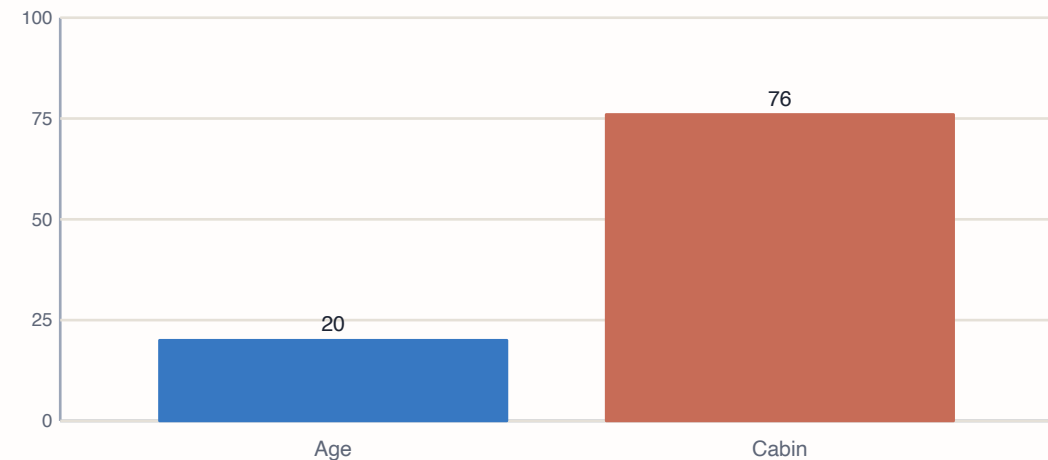
Titanic example: start with missing data

You cannot model survival well until you understand who is on the ship.

Step 1 — Catching the missing data

- Age is missing for about 20% of passengers.
- Cabin is missing for more than 75% of passengers.
- Deleting rows with missing age would discard a fifth of the sample.
- A better strategy is to impute age using titles such as Mr., Mrs., or Master.
- Cabin may be unusable as a direct feature, but “cabin missing” can still carry class-related information.

Missing values by variable

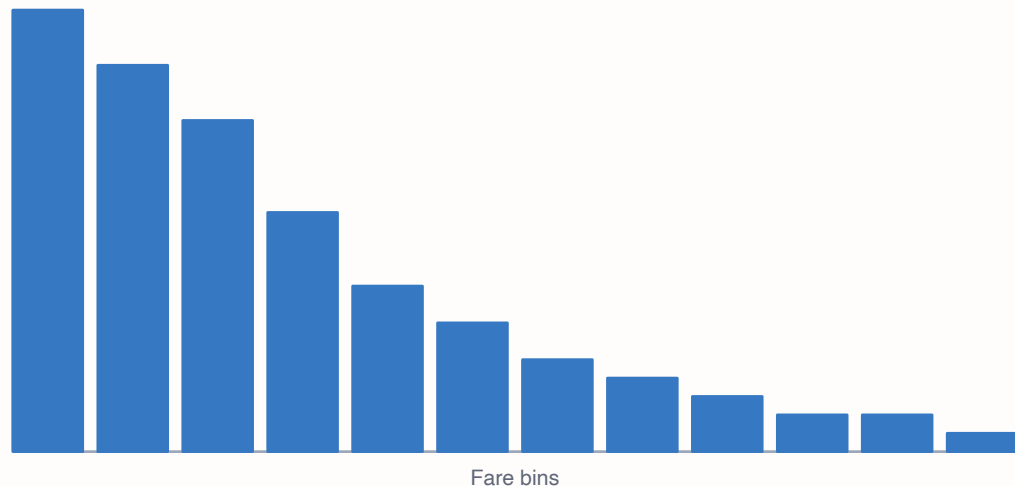


Insight: EDA changes the cleaning plan before modeling begins.

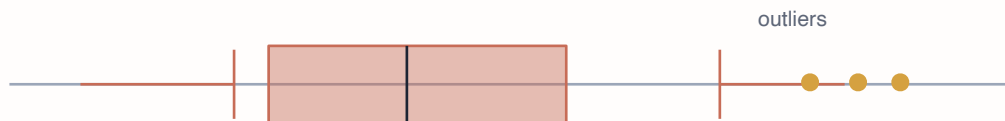
Titanic example: univariate analysis exposes outliers

The Fare variable is the key example in the course text.

Histogram of Fare



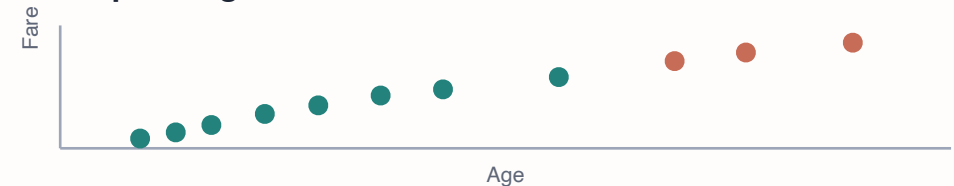
Boxplot of Fare



Step 2 — Hunting outliers

- The median fare is around \$14.
- The distribution is severely right-skewed.
- A few passengers paid more than \$500.
- If we later fit a linear model, those extreme fares will pull the regression line.
- That is why the text recommends a log transformation of Fare rather than blindly deleting those records.

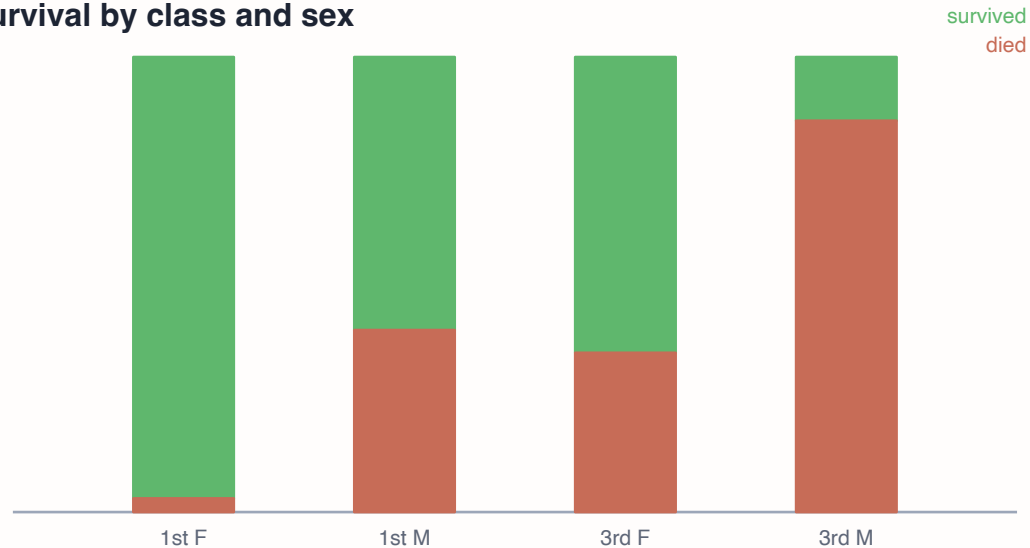
Conceptual Age vs Fare scatter



Titanic example: relationships reveal the strongest predictors

The key contrast in the text is survival by class and sex.

Survival by class and sex



Step 3 — Uncovering relationships

- A stacked bar chart makes the contrast immediate.
- 1st-class females are near a 100% survival rate.
- 3rd-class males have a very low survival rate.
- Before a single line of machine-learning code is written, EDA has already surfaced the most predictive features.

Feature-signal heatmap



Two ways to do EDA today

The methodology has changed dramatically in just a few years.

Traditional method: manual scripting

- Tooling: an IDE plus R or Python libraries such as tidyverse, ggplot2, pandas, and seaborn.
- Workflow: analysts manually write commands, summary statistics, loops, and correlation checks.
- Paradigm: if you forget to ask a statistical question, that relationship remains invisible.

Modern method: agentic discovery

- Tooling: LLMs with code execution, such as ChatGPT Advanced Data Analysis or Julius AI.
- Workflow: upload the raw data, describe the task in natural language, and let the system write and run the code.
- Paradigm: the process becomes conversational and iterative while the human directs the next question.

Critique of methods

Each approach fails in a different way if we stop thinking critically.

Traditional approach: confirmation bias

- Writing dozens of custom plots is tedious.
- Because of that effort, analysts tend to inspect relationships they already expect to find.
- That creates confirmation bias and can slide into p-hacking or visual pattern-hunting.

Agentic approach: the context deficit

- AI can flag mathematically unusual points without understanding domain history.
- A human knows that very expensive Titanic fares may reflect real luxury suites, not bad data.
- LLMs can also hallucinate: faulty code or a mislabeled axis can create a fabricated insight that looks convincing.

State of the art: multi-agent autonomous pipelines

The current frontier is not a single chatbot, but a coordinated analysis workflow.

Modern EDA is shifting from **one AI assistant** to **multiple interacting agents**.

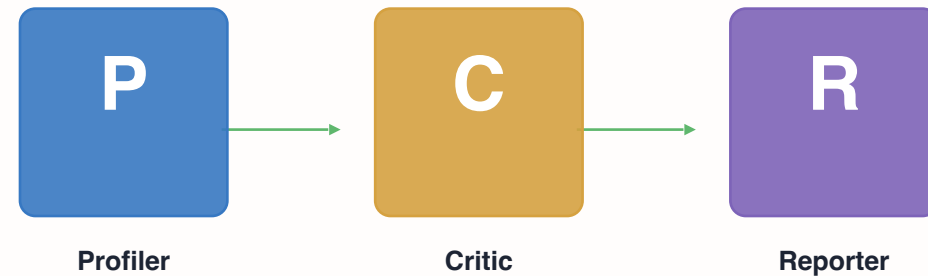
A **Profiler Agent** generates the first statistical picture of the dataset.

A **Critic Agent** reviews that output, challenges it, and requests deeper checks.

A **Reporter Agent** builds dashboards and summaries from the validated results.

The agents **exchange outputs, revise one another's work, and improve the analysis together**.

The human expert supervises the whole system and makes the final judgment.



Example output: feature signal matrix

	Class	Sex	Fare	Age
Survival	82	88	46	29

How AI was used in preparing this presentation

This keeps the assignment requirement, but makes the wording student-facing.

1. Prompts used

- A prompt asked for a step-by-step Titanic EDA focused on missingness, outliers, and bivariate relationships.
- A second prompt asked for a critical comparison between manual EDA and modern AI-agent exploration.

2. Raw output received

- The AI produced a strong overall structure.
- But it incorrectly referred to Titanic passengers as “patients.”
- It also suggested dropping the outliers too quickly.
And it added formulas that did not belong in an introductory EDA talk.

3. What we improved

- We corrected the terminology to “passengers.”
- We replaced deletion with a more defensible log transformation of Fare. We strengthened the critique by
- explicitly naming the “context deficit.” We want over too many iteration of this prestation because many errors

REMINDER

Assignment requirement

The final quiz requirement still needs to be included.

- ✓ Submit 4 multiple-choice questions with the creative presentation.
- ✓ Those questions should test understanding, not just memorization.
- ✓ The instructor will add 1 additional question.
- ✓ The class completes a short quiz during the dedicated quiz week.